

Lesson 3: Language of Probability

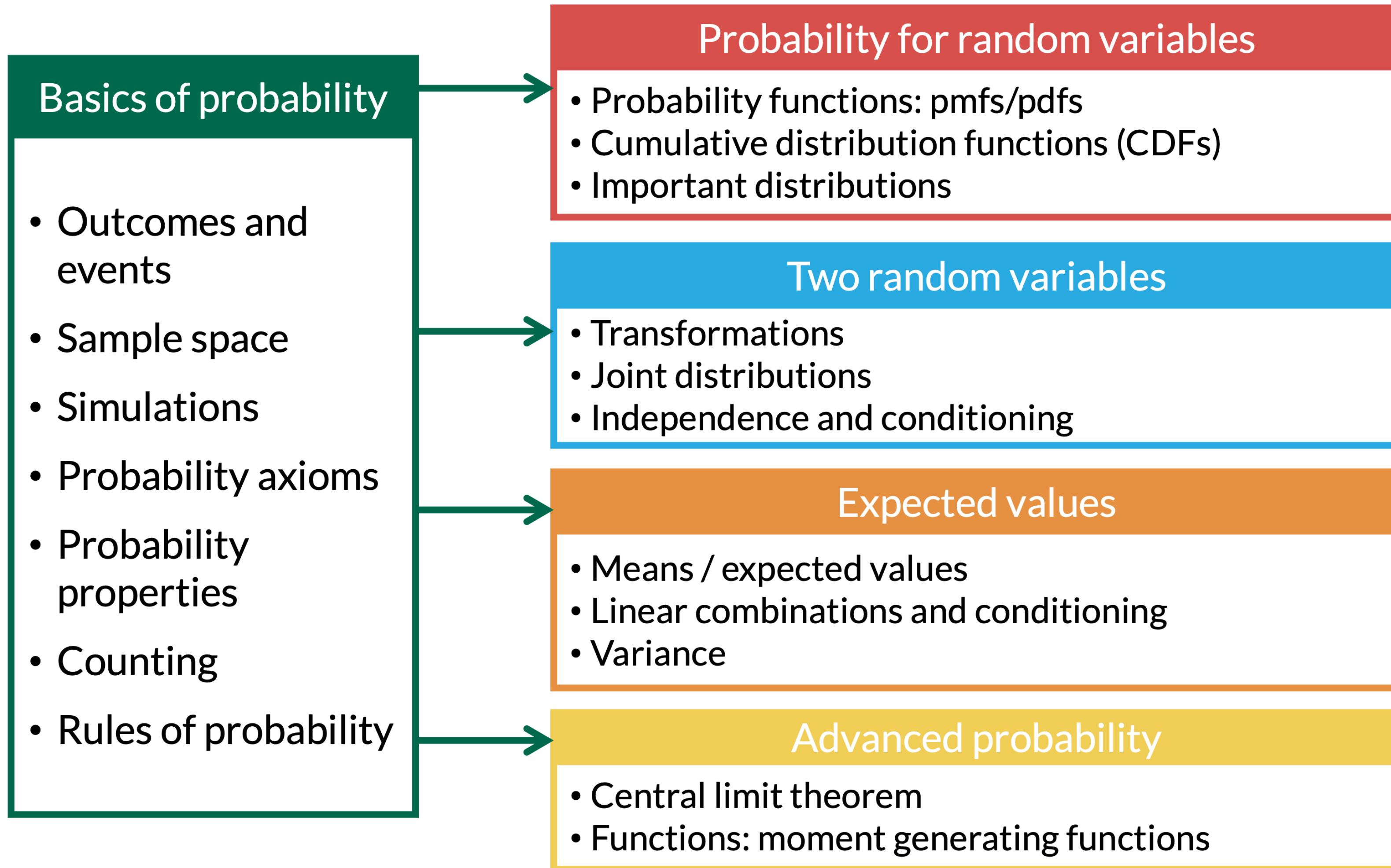
Nicky Wakim

2026-10-05

Learning Objectives

1. Use set notation, Venn diagrams, and the concepts of unions, intersections, complements, and mutually exclusive events to represent and describe events.
2. Apply the axioms of probability and related properties to calculate probabilities and prove simple results.
3. Explain and use De Morgan's Laws to simplify and solve probability problems.
4. Connect partitions and all rules of probability to calculate probabilities.

Where are we?



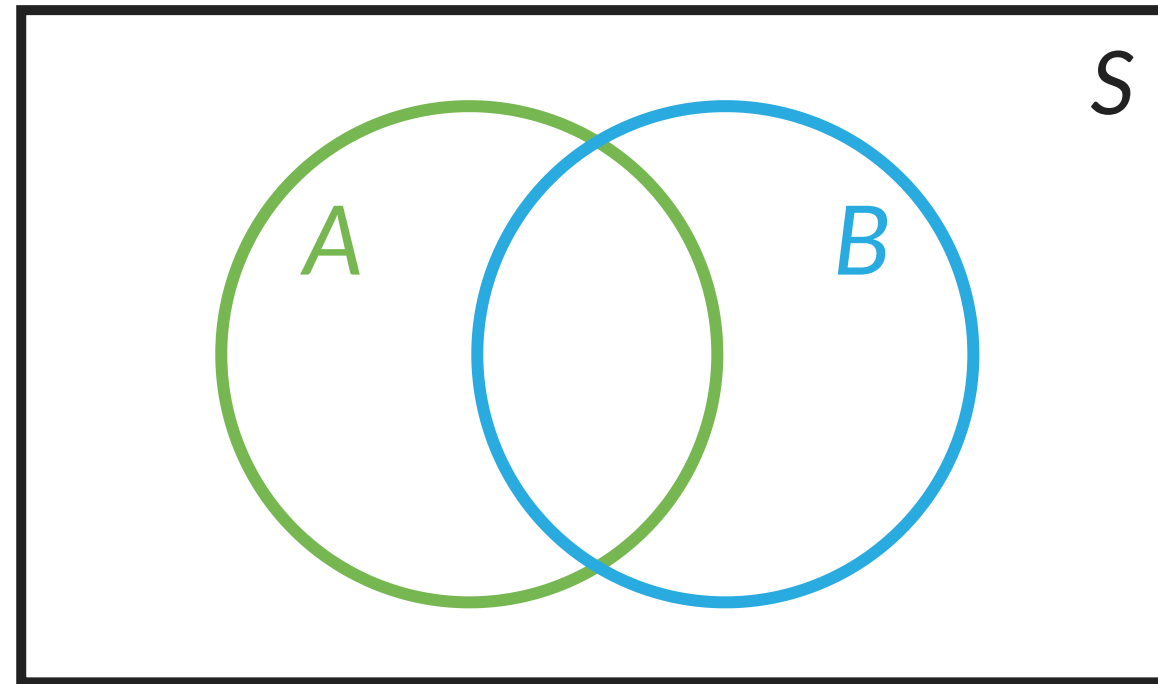
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Set Theory (1/2)

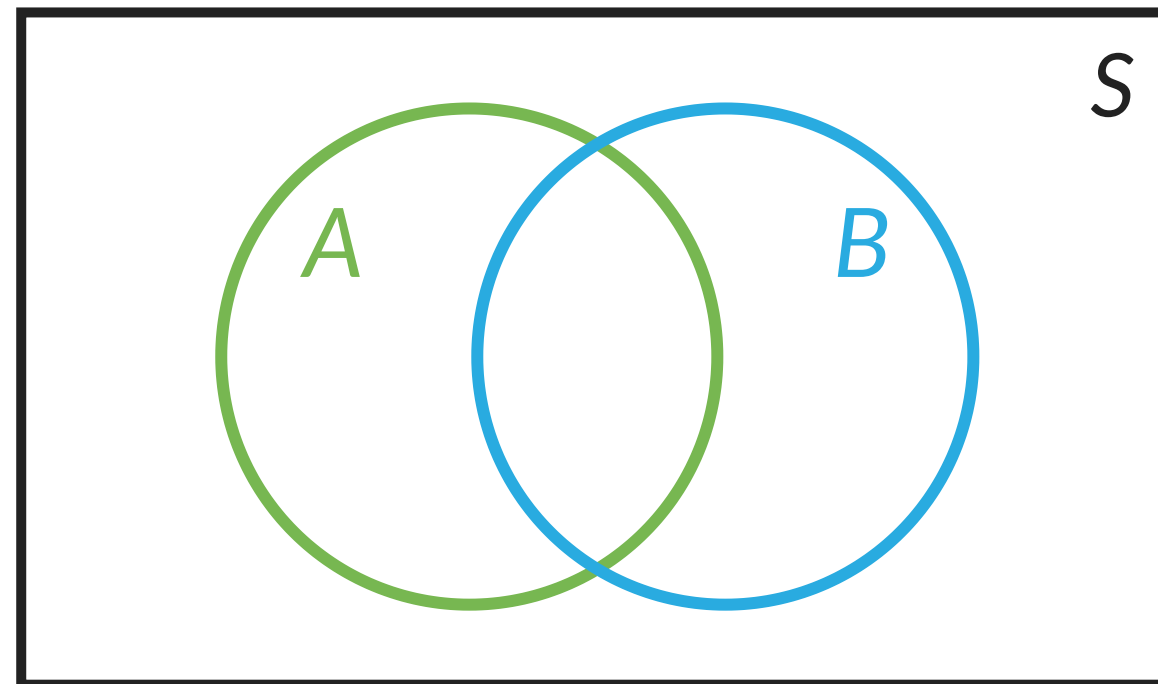
Definition: Union

The **union** of events A and B , denoted by $A \cup B$, contains all outcomes that are in A or B or both



Definition: Intersection

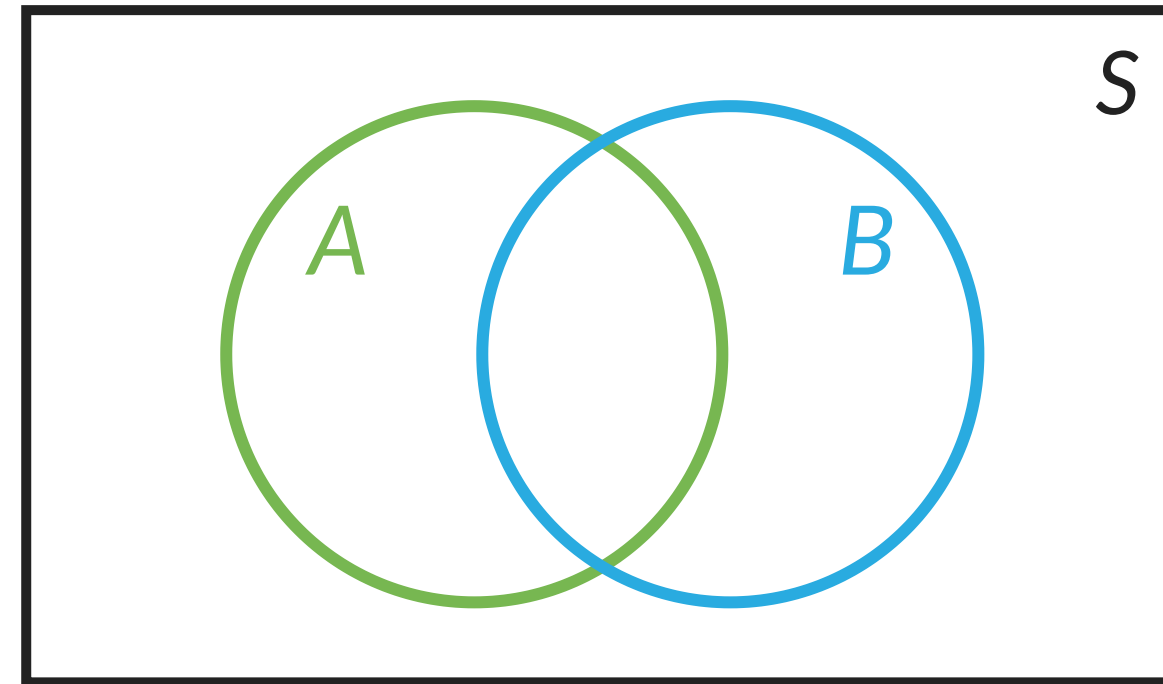
The **intersection** of events A and B , denoted by $A \cap B$, contains all outcomes that are both in A and B .



Set Theory (2/2)

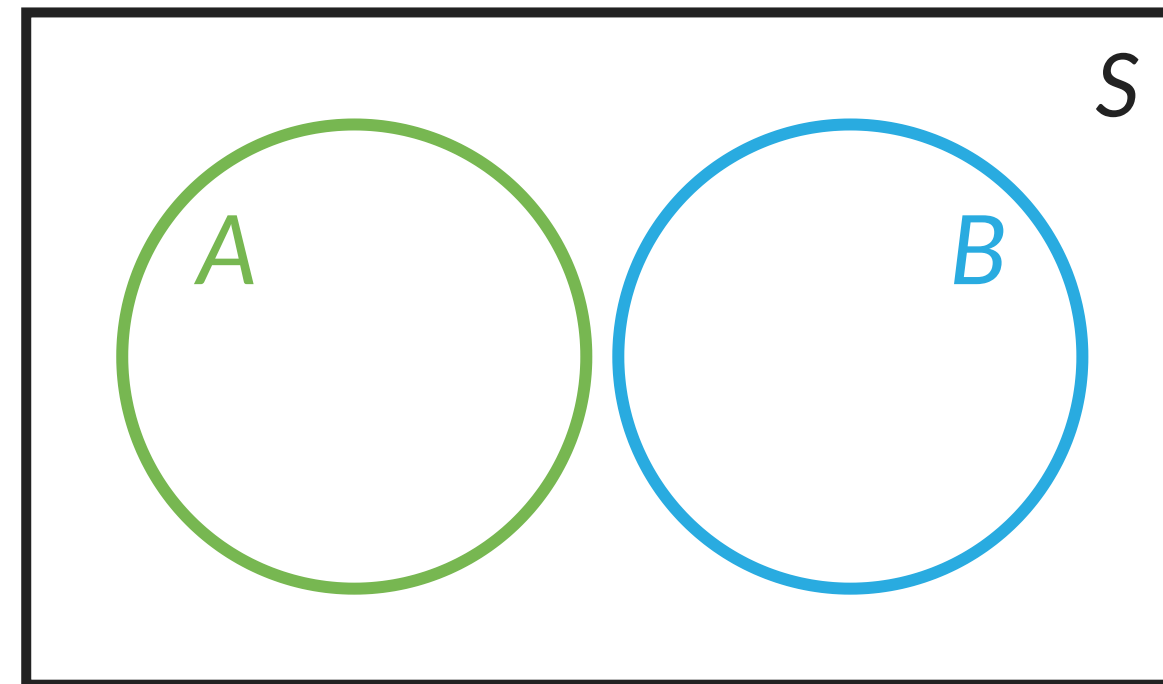
Definition: Complement

The **complement** of event A , denoted by A^C or A' , contains all outcomes in the sample space S that are *not* in A .



Definition: Mutually Exclusive

Events A and B are **mutually exclusive**, or disjoint, if they have no outcomes in common. In this case $A \cap B = \emptyset$, where $\emptyset$ is the empty set.



How can we code some of these? (1/4)

Example 1: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. **This time, let's say event A is rolling matching numbers, event B is rolling at least one 2, and event C is the sum of the dice is 5**

- First, we simulate rolling two four-sided dice. We start small, with only 5 simulations, so we can see every roll

```
1 set.seed(1022) ①
2 reps <- 5 ②
3 n_dice <- 2
4 rolls <- expand_grid(sim = 1:reps, die = 1:n_dice) |> ③
5   mutate(roll = sample(1:4, n(), replace = TRUE)) ④
```

1 rolls ①

```
# A tibble: 10 × 3
  sim die roll
<int> <int> <int>
1     1     1     4
2     1     2     3
3     2     1     2
4     2     2     1
5     3     1     1
6     3     2     4
7     4     1     2
8     4     2     2
9     5     1     3
10    5     2     3
```

How can we code some of these? (2/4)

Example 1: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. **This time, let's say event A is rolling matching numbers, event B is rolling at least one 2, and event C is the sum of the dice is 5**

- Now, we make one row per simulation, with a **TRUE/FALSE** column for each event

```
1 event_sims <- rolls |> ①
2   group_by(sim) |>      ②
3   summarise(A = n_distinct(roll) == 1, ③
4             B = any(roll == 2),        ④
5             C = sum(roll) == 5)        ⑤
```

```
1 event_sims ①
# A tibble: 5 × 4
  sim A      B      C
  <int> <lgl> <lgl> <lgl>
1     1 FALSE FALSE FALSE
2     2 FALSE  TRUE  FALSE
3     3 FALSE FALSE  TRUE
4     4  TRUE  TRUE  FALSE
5     5  TRUE FALSE  FALSE
```

How can we code some of these? (3/4)

Union

$$A \cup B$$

`A | B`

```
1 event_sims$A | event_sims$B ①
[1] FALSE TRUE FALSE TRUE TRUE
```

Intersection

$$A \cap B$$

`A & B`

```
1 event_sims$A & event_sims$B ①
[1] FALSE FALSE FALSE TRUE FALSE
```

Complement

$$A^c \text{ or } A'$$

`!A`

```
1 !event_sims$A ①
[1] TRUE TRUE TRUE FALSE FALSE
```

Mutually Exclusive

$$A \cap C = \emptyset$$

`any(A & C) is FALSE`

```
1 any(event_sims$A & event_sims$C) ①
[1] FALSE
```

How can we code some of these? (4/4)

Example 1: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. This time, let's say event A is rolling matching numbers, event B is rolling at least one 2, and event C is the sum of the dice is 5

- Same code, now with 10,000 simulations

```
1 reps <- 10000 ①
2 rolls <- expand_grid(sim = 1:reps, die = 1:4) ②
3 mutate(roll = sample(1:4, n(), replace = TRUE))
4 event_sims <- rolls |>
5   group_by(sim) |>
6   summarise(A = n_distinct(roll) == 1,
7             B = any(roll == 2),
8             C = sum(roll) == 5)
```

- Estimate the probabilities

```
1 event_sims |> ①
2   summarise(p_A = mean(A),
3             p_B = mean(B),
4             p_A_and_B = mean(A & B), ②
5             p_A_or_B = mean(A | B))
```

```
# A tibble: 1 × 4
  p_A    p_B p_A_and_B p_A_or_B
<dbl> <dbl>    <dbl>    <dbl>
1 0.255 0.437    0.0676    0.624
```

Exact: $\frac{4}{16} = 0.25$, $\frac{7}{16} = 0.4375$, $\frac{1}{16} = 0.0625$,
 $\frac{10}{16} = 0.625$

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Probability Axioms

Axiom 1

For every event A , $0 \leq \mathbb{P}(A) \leq 1$. Probability is between 0 and 1.

Axiom 2

For the sample space S , $\mathbb{P}(S) = 1$.

Axiom 3

If $A_1, A_2, A_3, \dots$, is a collection of **disjoint** events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The probability of at least one A_i is the sum of the individual probabilities of each.

Some probability properties

Using the Axioms, we can prove all other probability properties! Events A , B , and C are not necessarily disjoint!

Proposition 1

For any event A , $\mathbb{P}(A) = 1 - \mathbb{P}(A^C)$

Proposition 2

$\mathbb{P}(\emptyset) = 0$

Proposition 3

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$

Proposition 4

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

where A and B are not necessarily disjoint

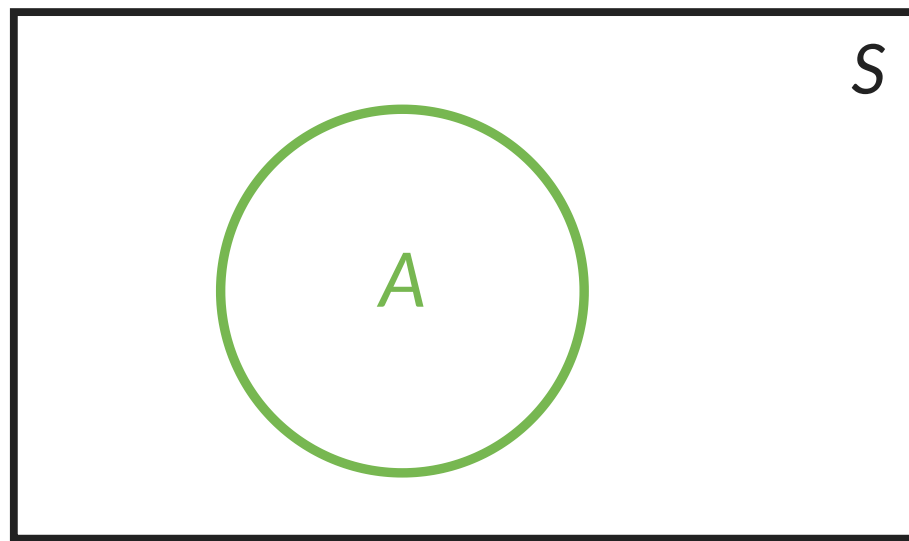
Proposition 5

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) = & \mathbb{P}(A) + \mathbb{P}(B) + \\ & \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \\ & \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C) \end{aligned}$$

Proposition 1 Proof

Proposition 1

For any event A , $\mathbb{P}(A) = 1 - \mathbb{P}(A^C)$



Use Axioms!

A1: $0 \leq \mathbb{P}(A) \leq 1$

A2: $\mathbb{P}(S) = 1$

A3: For disjoint A_i ,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 1 in our dice simulation

Proposition 1

For any event A , $\mathbb{P}(A) = 1 - \mathbb{P}(A^C)$

Two dice: B is “at least one 2”, so B^C is “no 2s”

```
1 event_sims |> ①  
2   summarise(p_B = mean(B),  
3             p_B_comp = mean(!B), ②  
4             p_B_prop_1= 1 - mean(!B)) ③
```

A tibble: 1 × 3

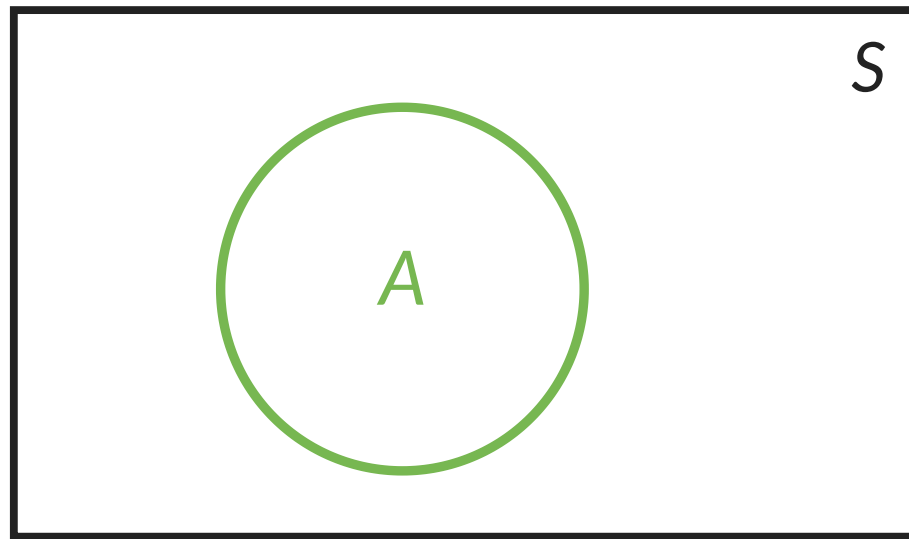
	p_B	p_B_comp	p_B_prop_1
	<dbl>	<dbl>	<dbl>
1	0.437	0.563	0.437

Exact: $1 - \mathbb{P}(B^C) = 1 - \left(\frac{3}{4}\right)^2 = \frac{7}{16} = 0.4375$

Proposition 2 Proof

Proposition 2

$$\mathbb{P}(\emptyset) = 0$$



Use Axioms!

$$\text{A1: } 0 \leq \mathbb{P}(A) \leq 1$$

$$\text{A2: } \mathbb{P}(S) = 1$$

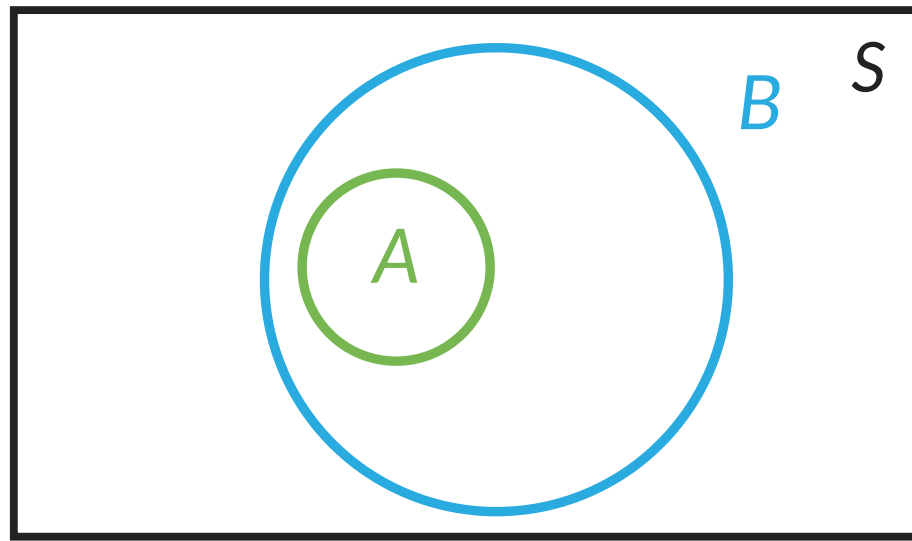
A3: For disjoint A_i ,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 3 Proof

Proposition 3

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$



Use Axioms!

A1: $0 \leq \mathbb{P}(A) \leq 1$

A2: $\mathbb{P}(S) = 1$

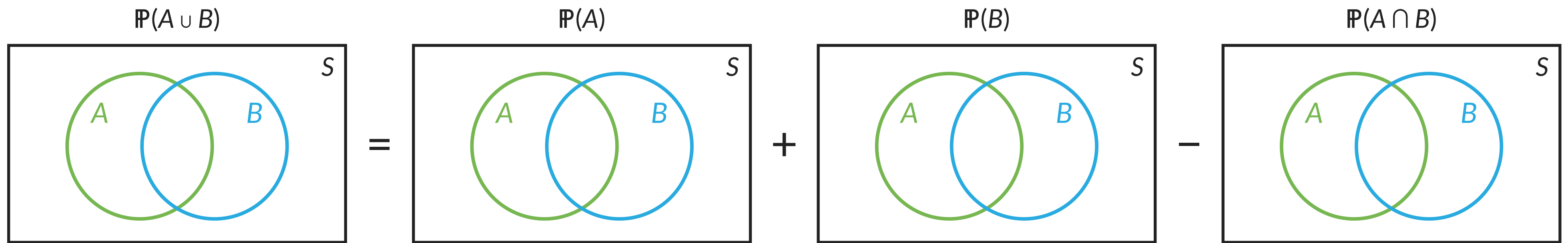
A3: For disjoint A_i ,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 4 Visual Proof

Proposition 4

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$



Proposition 4 in our dice simulation

Proposition 4

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

- A is matching numbers and B is at least one 2, so they overlap at (2, 2)

```
1 event_sims |>
2   summarise(p_A_or_B = mean(A | B),
3             add_subtract = mean(A) + mean(B) - mean(A & B))
```

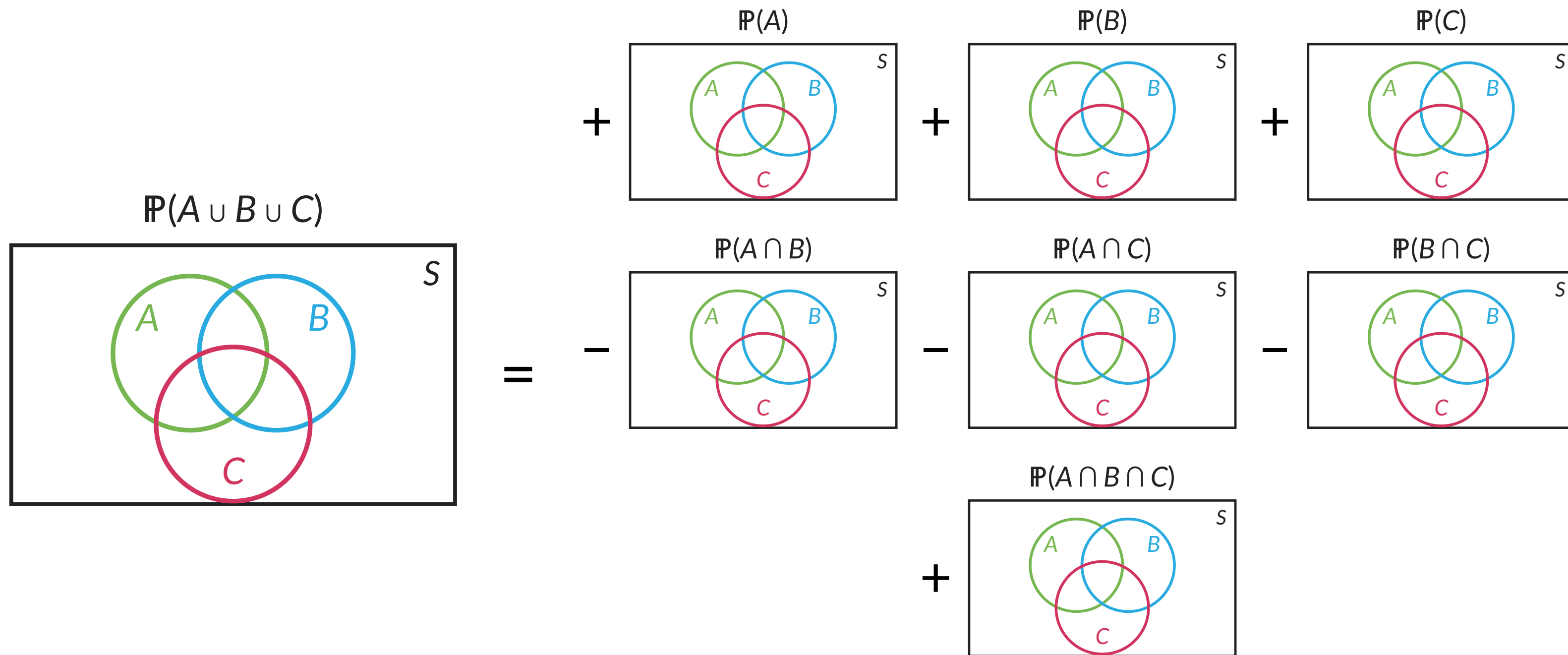
```
# A tibble: 1 × 2
  p_A_or_B add_subtract
  <dbl>      <dbl>
1  0.624      0.624
```

- Exact: $\frac{4}{16} + \frac{7}{16} - \frac{1}{16} = \frac{10}{16} = 0.625$
- The two columns match *exactly*, not just approximately

Proposition 5 Visual Proof

Proposition 5

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$$



Some final remarks on these propositions

- Notice how we spliced events into multiple **disjoint** events
 - It is often easier to work with disjoint events
- If we want to calculate the probability for one event, we may need to get creative with how we manipulate other events and the sample space
 - Helps us use any incomplete information we have

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De Morgan's Laws

Theorem: De Morgan's 1st Law

For a collection of events (sets) $A_1, A_2, A_3, \dots$

$$\bigcap_{i=1}^n A_i^C = \left(\bigcup_{i=1}^n A_i \right)^C$$

“all not A = (at least one event A)^C” or “intersection of the complements is the complement of the union”

Theorem: De Morgan's 2nd Law

For a collection of events (sets) $A_1, A_2, A_3, \dots$

$$\bigcup_{i=1}^n A_i^C = \left(\bigcap_{i=1}^n A_i \right)^C$$

“at least one event not A = (all A)^C” or “union of complements is complement of the intersection”

Example 2: Blood Pressures (BPs)

Suppose you have n subjects in a study. Let H_i be the event that person i has high BP, for $i = 1 \dots n$.

Use set theory notation to denote the following events:

1. Event subject i does not have high BP
2. Event all n subjects have high BP
3. Event at least one subject has high BP
4. Event all of them do not have high BP
5. Event at least one subject does not have high BP

Example 2: Blood Pressures (BPs)

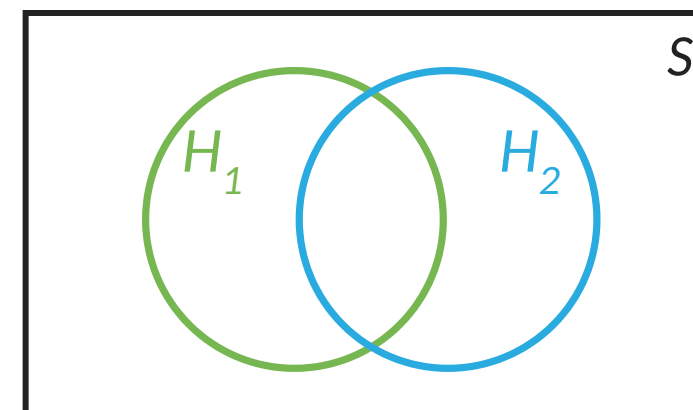
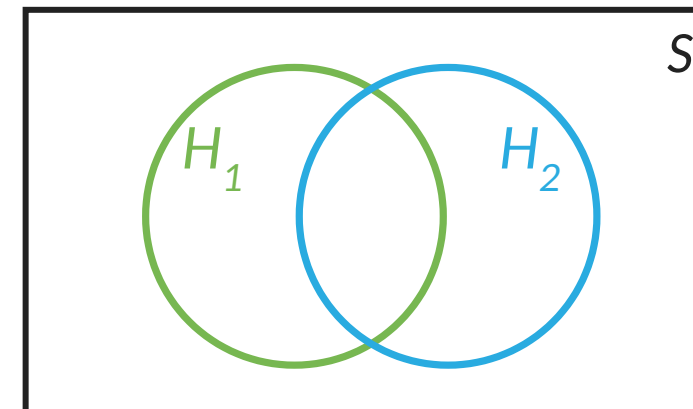
Suppose you have n subjects in a study. Let H_i be the event that person i has high BP, for $i = 1 \dots n$.

Use set theory notation to denote the following events:

1. Event subject i does not have high BP

2. Event all n subjects have high BP

3. Event at least one subject has high BP

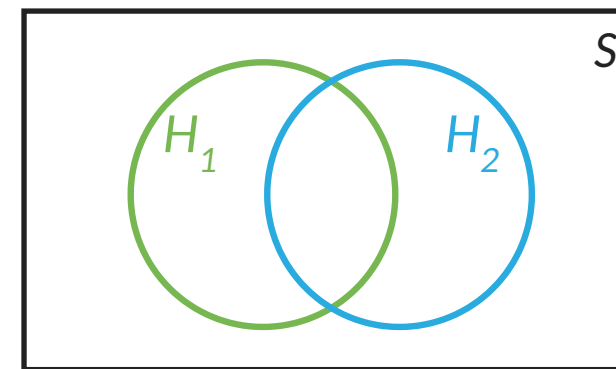


Example 2: Blood Pressures (BPs)

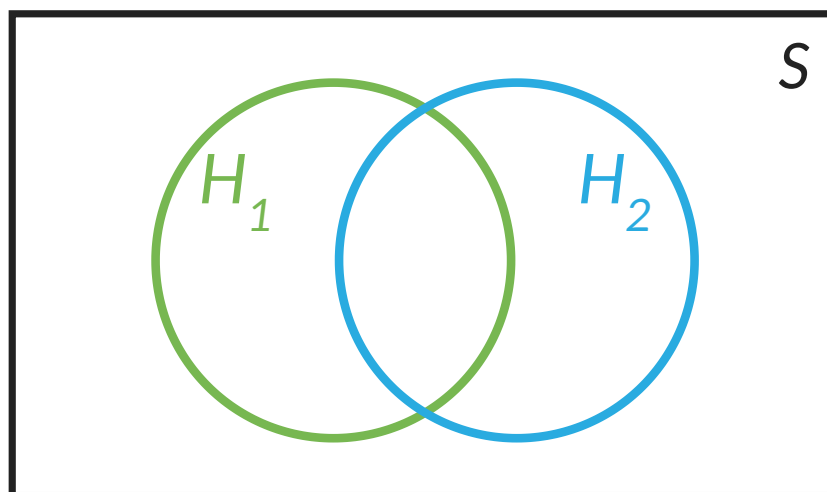
Suppose you have n subjects in a study. Let H_i be the event that person i has high BP, for $i = 1 \dots n$.

Use set theory notation to denote the following events:

4. Event all of them do not have high BP



5. Event at least one subject does not have high BP



De Morgan's Laws in simulation (1/2)

Simulate $n = 6$ subjects, each with a 30% chance of high BP, with 10,000 reps

```
1 n_subjects <- 6
2 p_high <- 0.3
3 subjects <- expand_grid(sim = 1:reps, subject = 1:n_subjects) |>
4   mutate(status = sample(c("high", "normal"), n(), replace = TRUE,
5                           prob = c(p_high, 1 - p_high)))
6 bp_sims <- subjects |>
7   group_by(sim) |>
8   summarise(all_not_high = all(status != "high"),      # Q2
9             not_any_high = !any(status == "high"),    # Q3
10            any_not_high = any(status != "high"),      # Q4
11            not_all_high = !all(status == "high"))    # Q5
12 head(bp_sims, 3)
```

A tibble: 3 × 5

	sim	all_not_high	not_any_high	any_not_high	not_all_high
	<int>	<lgl>	<lgl>	<lgl>	<lgl>
1	1	FALSE	FALSE	TRUE	TRUE
2	2	FALSE	FALSE	TRUE	TRUE
3	3	FALSE	FALSE	TRUE	TRUE

De Morgan's Laws in simulation (2/2)

- Check that each pair of columns agrees in every simulation

```
1 bp_sims |>
2   summarise(law_1_agree = mean(all_not_high == not_any_high),
3             law_2_agree = mean(any_not_high == not_all_high),
4             p_at_least_one_high = 1 - mean(all_not_high))
```

```
# A tibble: 1 × 3
  law_1_agree law_2_agree p_at_least_one_high
    <dbl>         <dbl>         <dbl>
1         1         1         0.876
```

- Exact: $1 - (0.7)^6 \approx 0.8824$
- We can find the probability of “all do not have high BP” more easily in the coming chapters

Remarks on De Morgan's Laws

- These laws also hold for infinite collections of events.
- Draw Venn diagrams to convince yourself that these are true!
- These laws are *very* useful when calculating probabilities.
 - This is because calculating the probability of the **intersection of events** is often much easier than the **union of events**.
 - This is not obvious right now, but we will see in the coming chapters why.

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Partitions

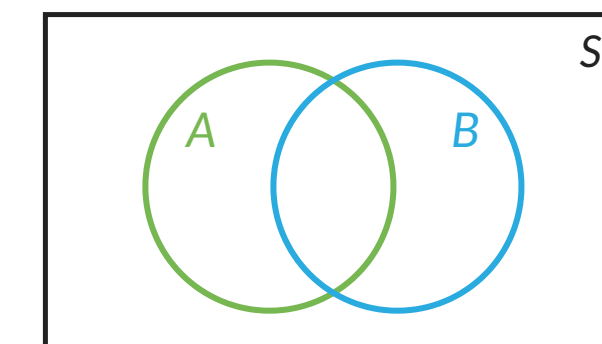
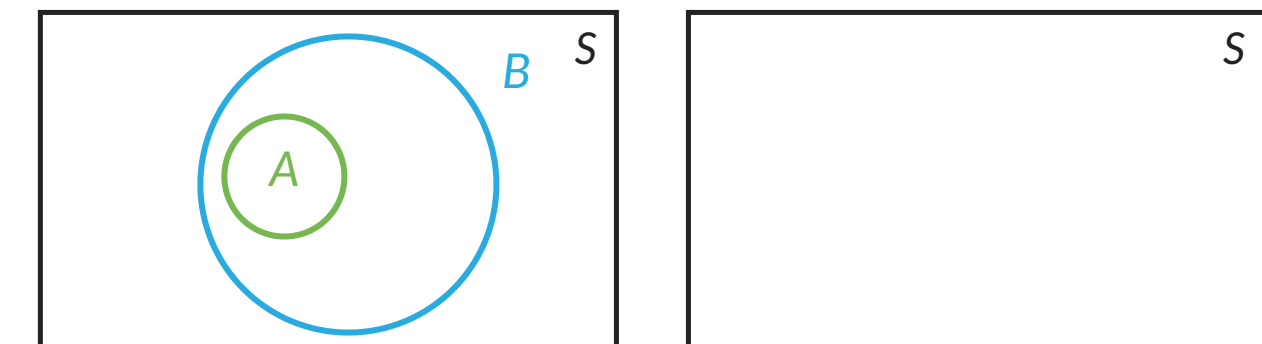
Definition: Partition

A set of events $\{A_i\}_{i=1}^n$ create a **partition** of A , if

- the A_i 's are disjoint (mutually exclusive) and
- $\bigcup_{i=1}^n A_i = A$

Note

- If $A \subset B$, then $\{A, B \cap A^C\}$ is a set of partitions of B .
- If $S = \bigcup_{i=1}^n A_i$, and the A_i 's are disjoint, then the A_i 's are a partition of the sample space.



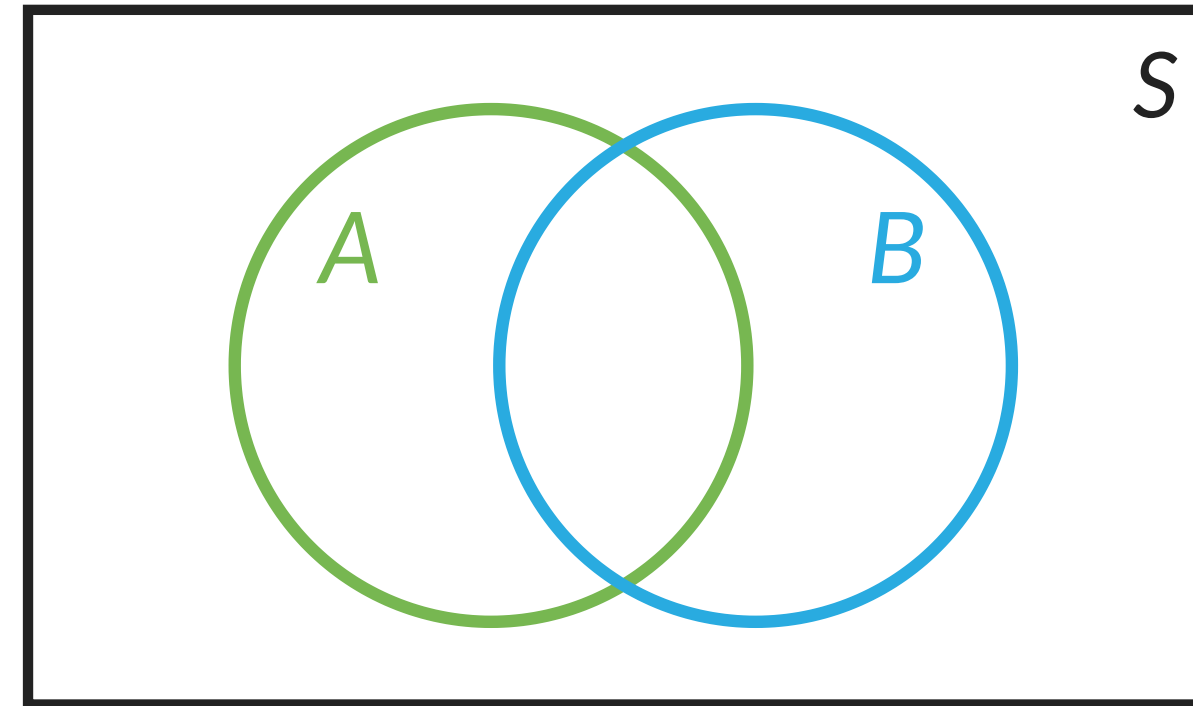
Creating partitions is sometimes used to help calculate probabilities, since by Axiom 3 we can add the probabilities of disjoint events.

Example 3: Weekly medications

If a subject has an

- 80% chance of taking their medication *this* week,
- 70% chance of taking their medication *next* week, and
- 10% chance of *not* taking their medication *either* week,

then find the probability of them taking their medication exactly one of the two weeks.



Hint: Draw a Venn diagram labelling each of the parts to find the probability.

Checking Example 3 with simulation

- After solving with the Venn diagram: simulate the four disjoint pieces (a partition of the sample space) and check the given facts

```
1 cells <- c("both", "this only", "next only", "neither") ①
2 med_sims <- expand_grid(sim = 1:reps) |> ②
3   mutate(cell = sample(cells, n(), replace = TRUE, ③
4             prob = c(0.6, 0.2, 0.1, 0.1)))
5 med_sims |>
6   summarise(p_this = mean(cell %in% c("both", "this only")), ④
7             p_next = mean(cell %in% c("both", "next only")),
8             p_neither = mean(cell == "neither"),
9             p_exactly_one = mean(cell %in% c("this only", "next only"))) ⑤

# A tibble: 1 × 4
  p_this p_next p_neither p_exactly_one
  <dbl>  <dbl>    <dbl>         <dbl>
1  0.797  0.696    0.105         0.297
```

- The first three columns should be close to the 0.8, 0.7, and 0.1 we were given. If they are, our partition is right, and the last column checks our answer.